This is only a note, based on Evans. And some of technique, which are mentioned in Evans, will be separated in many some topics of other notes.

For example: Mollification, we will discuss the theory in distribution theory.

Chapter 1 Introduction

1.1 PDE Problems

Definition 1.1.0.1 An expression of the form

$$F(D^k u(x), \dots, u(x), x) = 0 \qquad x \in U$$

Notice that $D^k u(x)$, thus

$$F : \mathbb{R}^{n^k + n^{k-1} + \cdots + 1} \times U \to \mathbb{R}$$

Its classification:

Definition 1.1.0.2

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-
-
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Definition 1.1.0.3

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PDES:

- Lienar PDEs:
- Nonlinear Systems;

1.1.1 Stragies

Some of PDEs:

- (1-Dimensional)
- (Wave)
- (Heat Core)
- (Schrödinger)
- (Burger's Equation)
- (Minimal Area)
- (Navier-Strokes' Equation)
  - ($\mu \neq 0$)
  - ($\mu = 0$,Euler Equation)

- (KortewegDe Vries (KdV) equation)

1.2 Warmeleitung Problem

Given:

- Region $\Omega \subseteq \mathbb{R}^3$;
- Masse $u : \Omega \times I \to \mathbb{R}$
- Flussschade $j : \Omega \times [0,T] \to \mathbb{R}^3$;

Prinzip

- $\forall x \in \Omega$,

$$\frac{d}{dt}\int_{B_r(x)} u(-,t)dy = -\int_{\partial B_r(x)} j \cdot v dS + \int_{B_r(x)} f(-.t)dy$$
$$\Rightarrow \int_{B_r(x)} (\partial_t u + div(j) - f)dy = 0$$
$$\Rightarrow \partial_t u + div(j) + f = 0$$

  Especially, u const$\Rightarrow$

1.3 Harmonic Equations

Definition 1.3.0.1 $\Omega \subseteq \mathbb{R}^n$ open. $u \in C^2(\Omega)$. u is harmonic$\Leftrightarrow -\Delta u = 0$

♣ **Example 1.3.0.2**

- $u(x) = Ax + b$;
- Quadrics $A \in \mathbb{R}^{n\times n}, u(x) := xAx^T = \sum_{i,j=1}^n x_i a_{ij} x_j \Rightarrow D^2u = A, \Delta u = tr(d^2u) = tr(A)$,so u is harmonic$\Leftrightarrow$ $tr(A) = 0$;
- Variables $n = 2$, $u(x_1, x_2) = f(x_1)g(x_2)$,

$$\Delta u =$$

- Holomorpihc Function $\Omega \subseteq \mathbb{C}$, harmonic$\Leftrightarrow$ u, v harmonic.

$$\Delta f = \Delta u + i\Delta v$$

1.4 Approxiamation through Convolution

Definition 1.4.0.1 $f, g : \mathbb{R}^n \to \mathbb{R}$,

$$(f * g)(x) := \int_{\mathbb{R}^n} f(y-x)g(x)$$

◇ **Lemma 1.4.0.2** (Convolution Approxiamation) $f \in L^p(\mathbb{R}), 1 \le p \le \infty, g \in$

- ○ **Proof**
- $p = 1$:
- $p = \infty$:
- $p \in (1, \infty)$:

□

◇ **Lemma 1.4.0.3** $f \in L^1(\mathbb{R}^n)$, $f = 0$ in $\mathbb{R}^n$ or B for

$$g \in L^1_\infty(\mathbb{R}^n) \Rightarrow f * g \in$$

If

∘ **Proof** ... □

◇ **Lemma 1.4.0.4** Suppose (η) satandard Dirac-sequence, then

...

1.5 Regularity

◇ **Lemma 1.5.0.1** Given a mollification η, $\eta_r(x) := r^{-n}\eta(\ )$ $u \in C^2(\Omega)$ and $u(x) = \oint_\partial B dS \Rightarrow u(x) = (u * \eta_r)(x), \forall \bar{B}_r(x) \subseteq \Omega$

∘ **Proof** ... □

♠ **Proposition 1.5.0.2** $\Omega \subseteq \mathbb{R}^n$ open, $U \in C^0(\Omega)$. If $u(x) = \int_{B_r(x)} u dy$, then $u \in C^\infty(\Omega), \Delta u = 0$.

∘ **Proof** ... □

Chapter 2 Linear PDEs

2.1 Transport Equation

2.2 Heat Equation

2.3 Wave Equation

2.4 Laplace Equation

2.5 Sobolev Theory

2.6 Elliptic Equation

Definition 2.6.0.1 ...

Chapter 3 Nonlinear PDEs